

WILLOW CREEK, AB, Canada

WMO: 712520

Lat: 53.386N

Lon: 118.347W

Elev: 1370

StdP: 85.91

Time Zone: -7.00 (NAM)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-34.6	-30.5	-38.2	0.1	-34.4	-33.6	0.2	-30.1	4.5	-6.6	3.9	-7.3	0.2	0	0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	26.2	12.7	24.1	12.0	22.1	11.3	14.0	22.4	13.1	21.4	12.2	20.1	2.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	9.9	14.8	10.2	9.2	14.0	9.1	8.5	13.3	43.8	22.3	41.0	21.4	38.6	20.4	17.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.8	4.1	3.5	DB	-39.0	29.6	1.7	1.6	-40.3	30.8	-41.3	31.7	-42.3	32.7	-43.5	33.8	(4)
(5)			WB	-39.0	16.2	1.6	0.9	-40.1	16.9	-41.1	17.4	-41.9	17.9	-43.1	18.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.4	-10.2	-9.3	-5.2	0.5	5.5	9.2	11.9	11.2	7.0	1.3	-6.2	-12.1	(6)	
	DBStd	9.97	7.94	7.38	6.87	4.35	3.58	2.81	2.98	2.98	3.70	4.43	6.85	7.32	(7)	
	HDD10.0	3696	627	542	472	286	143	47	15	22	102	269	487	684	(8)	
	HDD18.3	6562	885	775	730	536	397	273	200	220	339	527	737	943	(9)	
	CDD10.0	176	0	0	0	0	4	24	73	61	13	1	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	249	0	0	0	0	3	22	120	81	23	0	0	0	(12)	
	CDH26.7	37	0	0	0	0	0	3	20	12	2	0	0	0	(13)	
(14) Wind	WSAvg	1.2	1.0	1.1	1.3	1.4	1.4	1.3	1.3	1.2	1.1	1.1	1.0	0.8	(14)	
(15) Precipitation	PrecAvg	642	44	29	37	33	58	88	88	70	52	49	51	44	(15)	
	PrecMax	806	87	53	63	64	110	142	178	110	98	80	102	74	(16)	
	PrecMin	492	9	10	10	15	31	27	42	36	19	25	17	21	(17)	
	PrecStd	68	19	11	13	12	17	27	33	21	19	15	18	12	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	8.3	13.2	18.6	23.5	26.4	29.1	28.4	26.1	19.6	11.3	5.7	(19)	
		MCWB	2.2	2.1	4.7	7.3	10.1	12.3	13.4	13.6	12.1	8.9	5.3	1.5	(20)	
	2%	DB	5.0	5.3	9.1	14.6	20.8	22.8	26.5	25.7	23.1	15.1	7.0	2.9	(21)	
		MCWB	1.0	0.3	2.6	5.7	8.9	11.2	12.8	13.1	11.4	7.3	3.0	0.0	(22)	
	5%	DB	2.7	3.3	6.7	11.5	18.0	20.5	24.3	23.3	20.3	11.7	4.2	0.3	(23)	
		MCWB	-0.4	-0.9	1.1	3.9	8.5	10.4	12.4	12.3	10.4	5.7	1.2	-1.6	(24)	
	10%	DB	0.4	1.2	4.6	9.1	15.4	18.2	21.8	21.0	16.5	8.8	1.9	-1.8	(25)	
		MCWB	-1.7	-2.0	-0.2	2.8	7.2	9.4	11.7	11.7	9.0	4.0	-0.1	-3.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	2.6	4.8	7.8	11.3	13.7	15.2	15.5	13.4	9.5	5.6	2.2	(27)	
		MCDB	6.1	7.1	12.4	17.4	19.1	21.3	23.5	22.5	22.4	17.9	9.9	5.0	(28)	
	2%	WB	1.2	0.7	3.0	6.1	9.9	12.4	13.9	14.3	12.0	7.7	3.3	0.2	(29)	
		MCDB	4.4	4.6	8.0	13.3	18.3	20.2	22.3	22.3	21.3	13.5	6.5	2.3	(30)	
	5%	WB	-0.1	-0.5	1.5	4.5	8.8	11.2	13.0	13.2	10.8	6.2	1.6	-1.5	(31)	
		MCDB	2.6	2.8	5.9	10.6	16.7	18.5	22.0	21.1	19.2	11.1	3.7	0.2	(32)	
	10%	WB	-1.7	-1.9	0.3	3.1	7.6	10.1	12.2	12.2	9.4	4.5	0.1	-3.1	(33)	
		MCDB	0.3	1.0	4.0	8.4	14.8	16.8	20.3	19.5	16.0	8.1	1.5	-1.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.8	15.9	15.5	13.8	14.6	14.9	16.1	16.6	15.6	13.2	12.1	13.2	(35)	
		MCDBR	13.1	15.9	16.7	18.5	21.3	20.6	22.7	22.7	24.1	18.2	12.5	11.9	(36)	
	5% WB	MCWBR	9.4	11.3	10.6	10.2	10.9	10.0	10.4	11.2	12.6	10.7	8.7	9.3	(37)	
		MCWBR	12.1	15.0	15.5	16.9	19.1	17.6	19.1	19.6	21.6	16.8	11.4	11.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.240	0.245	0.280	0.310	0.331	0.346	0.336	0.337	0.303	0.278	0.246	0.232	(40)	
		taud	2.431	2.486	2.391	2.333	2.385	2.392	2.462	2.448	2.553	2.574	2.481	2.364	(41)	
	Ebn at Noon		783	898	917	924	918	903	907	886	885	842	769	713	(42)	
		Edh at Noon	51	66	91	110	111	112	102	98	78	61	48	44	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.79	1.64	2.73	4.21	4.94	5.10	5.30	4.51	3.14	1.75	0.84	0.61	(44)	
	RadStd		0.07	0.17	0.23	0.32	0.44	0.43	0.44	0.29	0.37	0.15	0.09	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page